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Research Statement

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1. BLACK HOLE MASS SCALING RELATIONS

1.1. *The $M_\bullet$–ϕ Relation*

Seigar et al. (2008) first presented evidence of a relationship between the mass of a supermassive black hole, ($M_\bullet$) and the pitch angle (ϕ) of its host galaxy. This relation introduced a powerful tool to estimate the masses of black holes in distant galaxies using only simple imaging data, without the need for telescope-expensive spectral data. In Berrier et al. (2013), we increased the number of spiral galaxies with directly measured supermassive black holes (SMBHs) from five (as it was in Seigar et al. 2008), to 22. In Davis et al. (2014), we generated a black hole mass function (BHMF) for local spiral galaxies by applying the updated $M_\bullet$–ϕ relation to a volume-limited sample of 144 galaxies from the Carnegie-Irvine Galaxy Survey (Ho et al. 2011). In a follow-up study (Mutlu-Pakdil et al. 2016), we also derived the BHMF for the complementary sample of 68 early-type galaxies. In Davis et al. (2017), we doubled the number of directly measured black holes used to generate the $M_\bullet$–ϕ relation (see Figure 1). This work received favorable coverage in the media, appearing in no less than 44 outlets,[1] including one magazine article I was requested to write for *Australasian Science* (Davis 2017).

1.2. *The $M_\bullet$–$M_{\star,\mathrm{sph}}$ Relation*

My main attraction to moving to Australia was the opportunity to work with Alister Graham. I consider him the world's foremost expert when it comes to the multicomponent decomposition of galactic light profiles and the modeling of their spheroids with Sérsic profiles. I have learned much from him during my time at Swinburne, becoming adept at the aforementioned skills and cutting-edge software (Ciambur 2015, 2016) that our research group utilizes. The majority of studies, particularly large surveys and pipelines, perform simple, automated, two-component (i.e., bulge/disk) decompositions to galaxies. Our research shows that this näiveté and oversimplification can be harmfully misleading, and is often wrong.

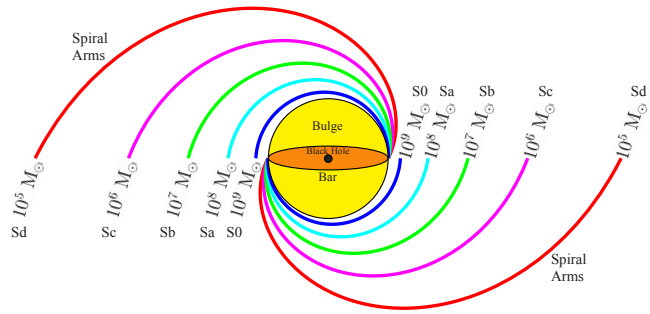

Figure 1. Spiral galaxy arms with varying degrees of tightness, and the corresponding galaxy morphological type and central black hole mass in units of our Sun's mass. This template can be used to estimate the black hole masses in spiral galaxies. The outermost spiral (——) has $|\phi| = 26\overset{\circ}{.}7$, which is indicative of a central black hole with a mass of $10^5\ M_\odot$ via Equation 8 from Davis et al. (2017).

I performed meticulous multicomponent decompositions of the spiral galaxy sample from Davis et al. (2017). In Davis et al. (2019a), we presented these detailed decompositions and our unprecedentedly accurate stellar bulge masses ($M_{\star,\mathrm{sph}}$). From our resulting sample, we performed a custom, sophisticated Bayesian linear regression to define the $M_\bullet$–$M_{\star,\mathrm{sph}}$ relation for spiral galaxies, such that $M_\bullet \propto M_{\star,\mathrm{sph}}^{2.44\pm0.33}$. This slope is significant due to its marked difference from the slope we later found for early-type galaxies, $M_\bullet \propto M_{\star,\mathrm{sph}}^{1.27\pm0.07}$ (Sahu et al. 2019a). Since the latter relation is almost ubiquitously reproduced in large N-body galaxy simulations for *all* types of galaxies, our research has important implications for such simulations; simulations that tune their results to a universal linear rate of growth between bulges and black holes need revision.

Furthermore, in Sahu et al. (2019a, see Figure 8) we showed that the seemingly sufficient single relation for early-type galaxies is misleading by revealing that early-type galaxies with and without a disk define two separate (parallel) $M_\bullet \propto M_{\star,\mathrm{sph}}^{1.9\pm0.2}$ relations, which are offset

[1] https://bendavis007.github.io/news#Davis:2017

by more than an order of magnitude (1.12 dex) in the $M_\bullet$-direction.

1.3. *The $M_\bullet$–$M_{\star,\rm tot}$ and $M_\bullet$–$M_{\star,\rm disk}$ Relations*

We released our study of bulge stellar masses in tandem with our study of total ($M_{\star,\rm tot}$) and disk ($M_{\star,\rm disk}$) stellar masses Davis et al. (2018), where we presented evidence that the $M_\bullet$–$M_{\star,\rm tot}$ relation is steeper, but not significantly less accurate than the $M_\bullet$–$M_{\star,\rm sph}$ relation for spiral galaxies. This flexibility is advantageous because it implies that black holes can be reliably predicted by measuring $M_{\star,\rm tot}$, a task that is significantly easier than isolating the stellar mass of its bulge. Saving time is appealing for extensive surveys and pipelines, for whom, producing thousands of time-consuming decompositions would be prohibitively time-intensive. We also presented evidence that $M_\bullet$ does weakly correlate with $M_{\star,\rm disk}$, a correlation which was previously thought not to exist (e.g., Kormendy & Gebhardt 2001).

1.4. *The $M_\bullet$–σ and L–σ Relations*

In our most accepted paper (Sahu et al. 2019b), we followed-up on the paradigm-shifting discovery in Sahu et al. (2019a) and searched for similar hidden substructures in the $M_\bullet$–(σ) central stellar velocity dispersion diagram. We investigated divisions based on the presence of a depleted stellar core (major dry-merger), a disk (minor wet/dry-merger, gas accretion), or a bar (evolved unstable disk), in a sample of 145 galaxies. Ultimately, we showed that Sérsic and core-Sérsic galaxies define two distinct relations: $M_\bullet \propto \sigma^{5.75\pm0.34}$ and $M_\bullet \propto \sigma^{8.64\pm1.10}$, with $\Delta_{rms|\bullet} = 0.55$ and 0.46 dex, respectively (see Sahu et al. 2019b, Figure 4). We also report on the consistency with the slopes and bends in the galaxy luminosity (L)–σ relations. Our new, type-dependent, $M_\bullet$–σ relations more precisely estimate $M_\bullet$ in other galaxies, and hold implications for galaxy/BH coevolution theories, simulations, feedback, and can improve predictions for the detection of long-wavelength gravitational waves using pulsar timing arrays and space-based interferometers.

1.5. *The $M_\bullet$–$v_{\rm max}$ and $M_\bullet$–$M_{\rm DM}$ Relations*

Some of my research involves investigating the connection between spiral galaxy structure, black hole mass, and dark matter halo mass ($M_{\rm DM}$). I have previous experience working with dark matter halo masses from my work on Seigar et al. (2014). Kennicutt (1981) first presented evidence that ϕ correlates with the maximum rotational velocity ($v_{\rm max}$) of disk galaxies. Ferrarese (2002) initially showed that black hole mass correlates with $M_{\rm DM}$. In Davis et al. (2019b), we updated both of these relations and demonstrated that these relations are consistent with our $M_\bullet$–$M_{\star,\rm tot}$ relation (Davis et al. 2018) and the Tully-Fisher relation (Tully & Fisher 1977).

2. INTERMEDIATE-MASS BLACK HOLES

Astronomers have discovered two sizes of black holes in the universe: SMBHs ($\geq 10^5\,M_\odot$) and solar mass black holes ($< 10^2\,M_\odot$). However, the so-called intermediate-mass black holes (IMBHs) theorized to fill in the evolutionary gap ($10^2\,M_\odot \leq M_\bullet < 10^5\,M_\odot$), remain elusive. The discovery of IMBHs would be a great boon to research/theory and provide potential multi-messenger targets for future generations of gravitational wave detectors.

Recently, I was involved with an international collaboration to use multiple scaling relations, including X-ray and radio luminosity relations, to search for possible IMBHs (Koliopanos et al. 2017). Subsequent research has included a *Chandra X-ray Observatory* Large Program (Soria 2016), which has imaged 52 galaxies with the ACIS-S detector. In Graham et al. (2019), we predicted the central black hole masses in these galaxies using the latest black hole scaling relations involving ϕ, σ, and $M_{\star,\rm tot}$; focused study of promising candidates is currently underway. We have bolstered this sample by using suitably deep archival *Chandra* images for an additional 22 spiral galaxies in the Virgo cluster, providing a complete sample of 74 spiral galaxies with star-formation rates $> 0.3\,M_\odot\,{\rm yr}^{-1}$. So far, this project has demonstrated the potential for half-a-dozen, high-impact papers.

Moreover, we made a serendipitous discovery of a potential shredded offset nuclear star cluster with an IMBH, identified by an X-ray source and optical/infrared counterpart. We have proposed for time on the Very Large Telescope (VLT) at the European Southern Observatory to collect adaptive optics (AO)-assisted integral field unit (IFU) spectra for this target using the Multi Unit Spectroscopic Explorer (MUSE) instrument **(I am the principal investigator)**. A second proposal for time at the Keck Observatory, using the Echellette Spectrograph and Imager to obtain AO-assisted spectra for 14 additional IMBH candidate targets, will be pursued.

Early-stage research is also proceeding in collaboration with Bogdan Ciambur (Observatoire de Paris), involving a focused study of 40–50 IMBH-candidate galaxies from (Graham & Scott 2013) and the bulgeless LEDA 87300 (Graham et al. 2016). Ongoing research includes measuring pitch angles for all of the later type spiral galaxies (i.e., Hubble type Sd) in the Third Ref-

erence Catalogue of Bright Galaxies (de Vaucouleurs et al. 1991) in a further search for IMBH candidates. For this work, I have assembled an international team, including: Alister Graham, Roberto Soria (University of the Chinese Academy of Sciences), Igor Karachentsev (Russian Academy of Sciences), Valentina Karachentseva (National Academy of Sciences of Ukraine), and Elena D'Onghia (University of Wisconsin). Additional multi-wavelength analyses are underway with the aid of Ryan Miller (Utica College).

3. LOGARITHMIC SPIRAL ARM PITCH ANGLE

Logarithmic spiral arm pitch angle was the focus of my Ph.D. thesis (Davis 2015) and continues to be a fundamental part of my research and a subject for which I possess a high level of expertise. Logarithmic spirals are ubiquitous throughout nature, manifesting themselves as optimum rates of radial growth for azimuthal winding in numerous structures such as mollusk shells, tropical cyclones[2], and the arms of spiral galaxies. See Section 2 from Davis et al. (2017) for a mathematical primer on ϕ. I have been personally involved in the development of three independent software packages to analyze images of galaxies and quantify their pitch angles:

1. The two-dimensional fast Fourier transform software, 2DFFT[3] (Davis et al. 2012, 2016), operates by decomposing galaxy images into logarithmic spirals and determining, for each number of harmonic modes ($1 \leq m \leq 6$), the pitch angle that maximizes the Fourier amplitude. This code was the first of its kind to measure ϕ as a function of radius.

2. The spiral template fitting software, SPIRALITY (Shields et al. 2015a,b), operates by examining pixel values on an image along logarithmic spiral coordinate axes with an origin at the center of a galaxy. It then computes the median pixel value along every axis. Spiral axes with ϕ that are drastically different from the real ϕ of a galaxy (or with the wrong chirality) will have little or no variance amongst different axes across the phase angle space of the spiral coordinate system. However, spiral axes that closely resemble the true ϕ of a galaxy will have increased variance as some axes will lie upon bright spiral arms and others will lie upon the darker gaps in between spiral arms. The actual pitch angle of the galaxy will be the set of axes with a pitch angle that maximizes the variance. This software is also quite useful for counting the number of spiral arms.

3. The computer vision software, SPARCFIRE[4] (Davis & Hayes 2014), uses machine learning to identify the clusters of pixels that form spirals in galaxies.

3.1. *Fundamental Planes*

Spiral density wave theory has existed and been the most popular theory governing spiral structure in disk galaxies, for more than a half-century now. Lin & Shu (1966) made a specific prediction that ϕ should be dictated by both the bulge mass and density of gas in the disk of a spiral galaxy. In Davis et al. (2015), we observationally confirmed this prediction.[5] Thus, the spiral density wave depends on the density of the disk, rather than the total mass of the disk, and it is the mass of the bulge that effectively anchors the spiral arm, a bit like setting the tension in a vibrating string of a violin by adjusting its tuning peg. Armed with increased data, my research group will continue to actively explore further trivariate relations between galactic parameters (e.g., van den Bosch 2016).

3.2. *Spiral Arm Theory Testing*

Spiral density wave theory predicts that when measured in different wavelengths of light, spiral arms will display slightly different pitch angles. In Pour-Imani et al. (2016) and Miller et al. (2019), we measured the pitch angles for spiral galaxies in multiple passbands from the far-ultraviolet to 8.0 μm. We found that ϕ was statistically tightest in 3.6 μm, and loosest in 8.0 μm images. I have been in talks about a collaboration with the PHANGS[6] research group about building upon these findings. I am interested in collaborating with with Tiantian Yuan, Guillermo Blanc (Carnegie Observatories), Eric Emsellem (ESO), Sharon Meidt (MPIA), Eva Schinnerer (MPIA), I-Ting Ho (MPIA), and Miguel Querejeta (ESO). I desire to lead an effort to directly measure spiral density waves observed in multiple wavelengths of electromagnetic radiation.

With the PHANGS group, I can utilize their high-resolution carbon monoxide imaging from the Atacama

[2] I have also measured the pitch angle of hurricanes, revealing a correlation between ϕ and a cyclone's barometric pressure. Ongoing research is proceeding with my colleague Patrick Treuthardt (North Carolina Museum of Natural Sciences), see https://github.com/treuthardt/Hurricanes.

[3] This software has since been updated to run in parallel, see https://github.com/treuthardt/P2DFFT.

[4] http://sparcfire.ics.uci.edu

[5] See the following link for an animated gif of the 3D relation, https://bendavis007.github.io/images/plane.gif.

[6] https://sites.google.com/view/phangs/home

Large Millimeter/submillimeter Array (ALMA) and Hα imaging from MUSE to measure the location of the spiral density wave in face-on, grand design spiral galaxies, with unparalleled precision. Recently, Peterken et al. (2019) presented evidence that pattern speed does not vary with radius, proving that a spiral density wave governs it. With the superior data of PHANGS, combined with pitch angle information, we should be able to surpass the quality of their test of density wave theory.

3.3. *Evolution of Spiral Galaxies*

Future work will focus on extending the local $M_\bullet$–ϕ relation to high redshifts. One of my colleagues, Tiantian Yuan (Swinburne), is an expert at source-plane reconstruction of gravitationally lensed high-redshift spiral galaxies. She discovered the most distant spiral galaxy to date, at $z = 2.54$ (Yuan et al. 2017). Tiantian's findings have opened up a new world of exploration for spiral galaxies. Just a few years ago, spiral galaxies were thought not to exist beyond $z = 2$, and the spirals that were found from that epoch were thought not to be "true" spirals, just conglomerations of clumpy gas clouds (e.g., Elmegreen & Elmegreen 2014).

Tiantian has compiled a list of over 20 spiral galaxies in the redshift bins of $z = 0.6$–1.2 and $z = 1.2$–1.8, plus a few at $z = 1.8$–2.5. She has AO-assisted, Keck IFU spectral data for some of these already, and hence rotational information. I am measuring pitch angles for these galaxies, and we are also artificially redshifting local spiral galaxies to similar redshifts as the high-redshift sample, from which we will draw comparisons. We intend to construct a high-z ϕ–$v_{\rm max}$ relation, and ϕ–$M_{\star,\rm tot}$ relation, and compare it with the local $z \sim 0$ relationships that we already have. We have initiated a collaboration with Elena D'Onghia to complement our research with her input and interpretation.

4. FUTURE PROJECTS AND PLANS

My research is ideally suited for graduate and undergraduate students. During my time as a graduate student, I mentored an REU student each summer, whom I taught my pitch angle software and how to measure spiral galaxies. Many of these summer projects led to these students co-authoring papers with me, including Matt Hartley, Luke Johns, and Russell Flatman. As a postdoc, I am an associate supervisor for Ph.D. students: Nandini Sahu and Suei-Hei (Dexter) Hon (investigating compact elliptical galaxies), and am mentoring an undergraduate honors student, James Sanders, who is quantifying the (X/peanut)-shaped structure in face-on disk galaxies (cf. Ciambur & Graham 2016). I have trained these students to use our decomposition software and am actively reviewing their projects and editing their manuscripts. Many of the students I have worked with have had limited experience with astronomy and possessed more general scientific backgrounds. With a little effort, I find it has been straightforward to train them in the methodologies necessary for my research and develop their knowledge as astronomers.

As ground-based detectors become more sensitive to longer wavelength gravitational radiation, they will be able to detect the coalescence of IMBHs from the mergers of late-type galaxies. My ongoing research into black hole mass scaling relations and BHMFs for spiral galaxies will help to establish estimates for the frequency of detections for the next generation of gravitational wave observatories and future space-based detectors like LISA. For example, Mapelli et al. (2012) discuss the detection of gravitational waves generated by the merging of a nuclear star cluster and an SMBH. More immediately, we can further improve the prediction of the amplitude and frequency for ground-based detections of long-wavelength gravitational waves produced by merging SMBHs using pulsar timing arrays (Shannon et al. 2015, Ryan Shannon is also at OzGrav).

Other avenues for funding/observing proposals will focus on one or more of my main projects: (1.) searching for IMBHs (§2), (2.) spiral arm theory testing (§3.2), and (3.) spiral galaxy evolution (§3.3),

1. I plan to conduct an exhaustive search for IMBH candidates in galaxies $\lesssim 10$ Mpc (Kraan-Korteweg & Tammann 1979), where follow-up observations from ALMA, with its 20 mas resolution, could directly observe the sphere-of-influence of an IMBH.

2. Building a collaboration with PHANGS, I will seek future time on ALMA and the VLT to explore spiral density waves in face-on spiral galaxies.

3. Legitimate high-redshift spiral galaxies at $z > 2$ constitute a brand new area of research. I want to expand my collaboration with Tiantian to include Deanne Fisher (Swinburne), who is an expert on turbulent, clumpy disk galaxies, and is the lead investigator of the Dynamics of Newly Assembled Massive Objects project (http://dynamo.swinburne.edu.au). Tying together these local analog galaxies with their high-z counterparts as the predecessors to spiral galaxies could yield intriguing insights on galaxy evolution.

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